CHRISTOPHER (CHRIS) VALLONE

MECHANICAL DESIGN ENGINEER | FLIGHT HARDWARE · ROBOTICS · TEST & VALIDATION

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ENGINEERING EXPERIENCE

SpaceX — Graduate Engineer | Starship Electronics · Hawthorne, CA — Aug 2026 – Present

- Performed mechanical design work on Starship flight electronics hardware, advancing and delivering priority engineering deliverables.
- Designed dedicated test hardware and developed multiple validation tests supporting flight hardware development.
- Conducted hardware testing, generating data that informed performance improvements.

Independent Robotics Hardware Engineer | Independent R&D — Jan 2023 – Present

- Designed custom BLDC robotic actuators with multi-stage planetary gearboxes, encoder feedback, and integrated mechanical packaging; developed a 300 Nm-class joint actuator prototype.
- Executed design–fabricate–test–redesign cycles across 100+ prototype units; built instrumented actuator test rigs, assembly jigs, and mounting fixtures to accelerate validation.
- Reduced actuator cost by 45% through DFM/DFA-focused redesign for low-cost legged-robot platforms.

Blood Shed Motors EV — Engineering Intern | Austin, TX — Jul 2019 – Jun 2020

- Designed and built a 400 V / 150 A battery test stand to support repeatable EV battery qualification testing.

SELECTED ENGINEERING PROJECTS

Bates Propulsion Labs Rocket Team — Test Stand & Electrical Lead | UT Dallas — Aug 2019 – May 2020

- Designed and welded a steel rocket-engine test frame; integrated a 24-sensor DAQ system with less than 3 ms latency to support hot-fire operations.

SynQor HV Rack-Mount Supply — Senior Design Capstone | UT Dallas — May 2022 – Dec 2022

- Designed a 3 kW isolated power supply, including a two-board, 400 V PCB layout in Altium and digital PID control over SPI.

EDUCATION

The University of Texas at Austin — M.S., Mechanical Engineering (in progress; expected 2028)

The University of Texas at Dallas — B.S., Electrical Engineering, 2022

TECHNICAL SKILLS

Mechanical Design: SolidWorks, Autodesk Inventor, Creo, GD&T, FEA, DFM/DFA, mechanisms, actuation

Fabrication & Test: CNC milling/turning, sheet metal, welding, rapid prototyping, fixtures, instrumentation, DAQ

Electrical & Controls: BLDC motors, encoders, CAN, Altium, PCB bring-up, HV electronics, STM32, BMS

Software & Tools: Python, C/C++, MATLAB, Git, LTspice; Autodesk Inventor Certified User, IPC-A-610